SUJAL SHRESTHA

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https://asur-sujalshrestha.github.io/sujal-portfolio/

SUMMARY

Backend Software Engineer experienced in building scalable, secure, and cloud-native applications using Java and Spring Boot. Skilled in Hexagonal Architecture, Spring Modulith, microservices, multi-tenancy, Keycloak, PostgreSQL, Redis, Elasticsearch, Docker, and Kubernetes. Experienced in developing real-time WebSocket solutions, system integrations, data synchronization, and enterprise security features.

WORK EXPERIENCE

Java Developer : Pexus Tech. / ING Tech **Dec 2025 - Present**

Enrollment V2:

- Designed and implemented Enrollment V2 using Spring Modulith and Hexagonal Architecture, establishing modular and maintainable domain boundaries.
- Implemented Keycloak-based IAM and permission-driven RBAC for secure authentication and fine-grained authorization.
- Designed schema-based multi-tenancy with PostgreSQL and implemented tenant-aware application functionality.
- Built cloud-native infrastructure using Docker and Kubernetes, with MinIO, Redis, Elasticsearch, Cloud Config, and HashiCorp Vault for storage, caching, logging, configuration, and secrets management.
- Implemented WebSocket-based real-time student admission events across application modules to enable instant updates.

Attendease:

- Contributed to a multi-tenant microservice architecture supporting isolated operations across multiple tenants.
- Implemented WebSocket-based routine import for synchronizing student routines across different groups and tenants.
- Integrated biometric attendance services with scheduled processing and synchronization of attendance logs.
- Developed email and report generation services and supported onboarding and configuration of new tenants.
- Migrated Keycloak authorization from role-based to permission-based authorities and integrated Attendease with Enrollment through a reverse proxy.

Authorization Server:

- Designed and implemented an Authorization Server using Spring Modulith and Hexagonal Architecture to provide centralized authorization capabilities.
- Integrated with Keycloak IAM to retrieve user details, roles, and granular permissions for fine-grained access control.
- Developed Feign Client-based integration to provide authorization services to both the Attendease and Enrollment systems.

SKILLS

- **Languages**: Java, SQL
- **Backend**: Spring Boot, Microservices, Spring Modulith, Hexagonal Architecture
- **Security**: Keycloak, IAM, OAuth 2.0/JWT
- **Databases**: PostgreSQL, MySQL, MS SQL
- **Infrastructure**: Docker, Kubernetes, CI/CD
- **Frontend**: React.js
- **Messaging & Real-time**: Kafka, RabbitMQ, WebSocket
- **Caching & logs:** Redis, Elasticsearch
- **Storage & Configuration:** MinIO, HashiCorp Vault, Cloud Config
- **Version Control:** Git, GitHub, GitLab

EDUCATION

Bachelor in Computing Science **Nov 2022 - june 2025**
Islington College

+2 level in Science **2020-2022**
Capital College (CCRC)

INDIVIDUAL PROJECTS

Online Learning and Collaboration Platform (Spring Boot, Postgres, React):

- Developed a full-stack collaboration platform with Spring Boot, React, and PostgreSQL, supporting organizations, classrooms, groups, assignments, and announcements.
- Implemented JWT-based authentication, role-based authorization, real-time WebSocket communication, and file/content sharing.

Hospital Management System (Java, Spring Boot, React, PostgreSQL, Microservices):

- Developed a microservices-based hospital management system for patient registration, doctor management, appointments, and medical records.
- Implemented secure authentication, role-based access control, REST APIs, and normalized PostgreSQL database design.

Online Food Ordering System (Java Spring Boot, React, PostgreSQL):

- Developed an end-to-end food ordering platform with restaurant, customer, admin, and delivery workflows.
- Integrated the Khalti payment gateway and implemented restaurant management, order tracking, rider assignment, and delivery management.

CERTIFICATIONS

- AWS Cloud foundation
- AWS Cloud Data Engineering
- AWS Cloud Machine Learning
